Full-Signal PNNN versus PN-GMP DOMP Comparison

Objective and methodology. Complex GMP, its exactly equivalent coupled PN-IQ representation, independent PN-IQ controls, and parameter-matched PNNNs use the same modeled-block X/Y samples. DOMP is used only during linear identification.

Fair comparison protocol. The classic selector provides 19661 identification rows (approximately 4% of the capture). An internal 16711/2950 split selects regularization and neural epochs. Every final model is fitted on all identification rows and evaluated on all 491520 signal rows. **The full-signal evaluation includes the identification samples and therefore must not be interpreted as an independent generalization test.**

Main results. NMSE uses $10\log_{10}(\sum|y-\hat{y}|^2/\sum|y|^2)$.

Model	Real parameters	Full-signal NMSE (dB)	FLOPs/sample	Additional operations/sample
Independent PN-IQ full	358	-38.534	1062	12 magnitude, 12 sqrt, 2 division, phase normalization/restoration arithmetic in FLOPs
Complex GMP DOMP parameter-matched	366	-38.176	1944	15 magnitude, 15 sqrt
PNNN N12 sparse	358	-35.900	876	14 magnitude, 14 sqrt, 2 division, 12 ELU, up to 12 exp, phase normalization/restoration arithmetic in FLOPs

Computational cost. FLOPs/sample counts real additions and multiplications required for one complex output, including feature generation, coefficients or weights, biases, phase normalization, and phase restoration. Magnitudes, roots, divisions, exponentials, and ELUs remain separate. DOMP is not an inference cost. The sparse-PNNN FLOP count assumes that zero weights are skipped. An implementation that executes the original full matrices will require 2252 FLOPs/sample instead.

Interpretation and limitations. Independent PN-IQ provides the strongest NMSE in the main comparison; the parameter-matched complex GMP uses the fewest real parameters; and the sparse PNNN uses the fewest counted FLOPs when zero weights are actually skipped, while adding ELU evaluations. Complex GMP DOMP-100 and coupled PN-IQ remain numerically equivalent with relative error $1.321e-11$. These analytical counts do not establish latency, energy, or FPGA resource superiority.